COEP TECHNOLOGICAL UNIVERSITY

PRESENTS

MINDSPARK'26

INFINITE CONTINUUM

RULEBOOK

3RD & 4TH October

www.mind-spark.org

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ROBOTICA

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BOT WRESTLING - - -

BOT WRESTLING

EVENT OBJECTIVE:

Bot Wrestling is a competitive event where two robots, controlled either wirelessly or via wired connections, face off with the objective of pushing the opposing robot out of the designated arena. These robots must be operated remotely operated, ensuring a battle of strategic manoeuvring and technological prowess.

EVENT STRUCTURE:

The arena features a single white ring as the boundary. The objective for each robot is to push the opposing robot entirely out of this platform. Points are awarded when a robot successfully forces its opponent completely outside the white ring.

Rounds:-

Qualifier Round:

1. Qualifier round will be held prior to the knockout round.
2. In each match, two teams will compete against each other for 5 rounds.
3. The teams will be ranked according to their points.
4. Top 16 teams will qualify for the finals. Other than this, the performance in the qualifier round is of no significance to the contest.

Knockout Round:

1. Two teams will compete against each other` same as the qualifier round.
2. The team with best of 7 rounds wins the match.
3. The team that pushes the opposition robot out of the arena is declared as the winner of that round.
4. The scoring pattern will be uniform throughout and the scores will be considered only in case of a tie i.e. if no team is able to push the opponent's robot out of the arena in that round.
5. In case of a tie in a round (i.e. none of the team's robot is able to push the opposition robot out of the arena), the winner will be decided by the points earned by each team in that round.
6. In case of a tie in the match, the winner of the match will be decided by a tiebreaker.

TASKS:

1. Teams must construct a robot that can be operated wirelessly via a remote controller.
2. The objective of the robot is to push the opposing team's robot out of the arena. The team that successfully does so wins the match.

ARENA:

1. The arena will be square in shape of side 2400 mm.
2. The main arena will be circular in shape, with a maximum diameter of 2200 mm.
3. A central white line, 25.40 mm in width, divides the arena into two equal halves.
4. White border squares measuring 300 mm x 300 mm are located on opposite sides of the central line, indicating the starting zones for the two competing teams.
5. The arena will be constructed from wood, with all markings, including the zones, painted onto the surface.

ROBOT SPECIFICATIONS:

1. The robot should fit in a square box of area 300mm x 300mm.
2. There is no height limit on the construction of the robot.
3. The combined weight of the robot and the remote must not exceed 5 kg.
4. If the team is using a laptop or a mobile phone for controlling the robot then its weight is need not to be mentioned.
5. The robot must be stable and must stand on its own at the beginning of the run when kept at the starting point. Robots not fulfilling this criterion will be disqualified.
6. During the match, the robot is allowed to extend or enlarge itself, provided it does not cause any damage to the arena. However, the robot must not leave any parts or materials behind while competing. Additionally, the robot should not scratch, damage, or destroy any part of the arena or its accompanying components. Failing to abide with above rules will lead to immediate disqualification.

POWER SUPPLY:

1. The robot must be entirely self-sufficient and require no external assistance. It must not use any energy produced through burning fuels that may have any potential risk of fire hazard.
2. The robot must have an internal on-board power source, external power supplies are not allowed.

3. The voltage within the robot must not exceed 12 V DC at any point.

POINT SYSTEM

1. One match shall consist of 5 rounds, within a total time of 5 minutes, unless extended by the judges
2. Each team earns 1 point for pushing the opposition outside the game zone.
3. If the robot moves outside on its own the opposition team will earn 1 point.
4. Negative score of a team will be considered valid.
5. Decision given by the judges shall be treated as final and binding to all.
6. The organizers reserve the rights to change any or all of the above rules as they seem fit.

HAND TOUCH:

1. No hand touches are allowed in between the matches unless both the operators of the respective robots want a restart.
2. If a restart is taken, then both the robots will be placed back in their respective starting zones.
3. Even if a participant takes a hand touch, the clock will continue to run and will not be reset as the robots start again from the starting points.

RULES:

1. Qualifier Matches: Each match consists best of 5 rounds.
2. Knockout Matches: Each match consists best of 7 rounds.
3. Top 16 teams selected for Knockout Matches will be selected on total scores in qualifier with total time in consideration.
4. Before the match 1-minute setting time will be given. The run time clock will commence after that 1-minute time limit even if the handler is still adjusting the robots.
5. 10 seconds setting time will be given in between the rounds.
6. The robot must start from the starting point.
7. The operator may abort the run at any time. In such a case, the opposite team will be declared as the winner.
8. No hand touches are allowed in between the matches unless both the operators of the respective robots want a restart.
9. Use of Jammers is not allowed. Participants found violating this rule will be disqualified from the competition immediately.
10. There can be interference problems related to RF modules in case they are being used. It is the responsibility of the participants to overcome the problems. Organizers will not be responsible for it.
11. Decision by the judges shall be treated as final and binding to all.

12. The organizing committee will ensure that fire extinguishers are readily available on-site to address any potential fire hazards during the event.
13. Participants are expected to maintain decorum. We reserve the due right to expel the candidate with inappropriate conduct.
14. In case of any disputes, the decision of event heads and judges will be final and binding to all.
15. Rules of the event are subject to change without prior intimation. Participants are requested to check the MindSpark'26 website (www.mind-spark.org) for regular updates.
16. **Teams are advised to bring one bot per competition. If the same bot is used for multiple competitions, no rescheduling or accommodation will be provided in case of scheduling conflicts.**
17. **The same team, or the same bot, with different team name is not permitted to participate multiple times in the same event. Any violation of this rule will result in immediate disqualification.**
18. **Teams are permitted to enter the same bot in different events only if significant modifications are made to adapt the bot to the requirements of each event. For example, a bot modified by replacing its pushing edge with a C-bracket may then qualify to participate in RoboSoccer.**

TEAM AND FEE STRUCTURE:

Team size : Maximum 4 participants per team.
Registration fee :₹ 499 per team.

EVENT HEAD:

Pranav Pawar: +91 96578 08045
Prathamesh Kothawale: +91 92090 67635

NOTE:

1. Prizes are subject to change depending on number of participants.
2. Registration fee is non-refundable.
3. Prize Money is subject to change.
4. Participants must report within their allotted time slot. Failure to do so will result in immediate disqualification from the event.
5. The Event Head reserves the right to make the final decision in all matters concerning the event.

ROBO SOCCER

EVENT OBJECTIVE

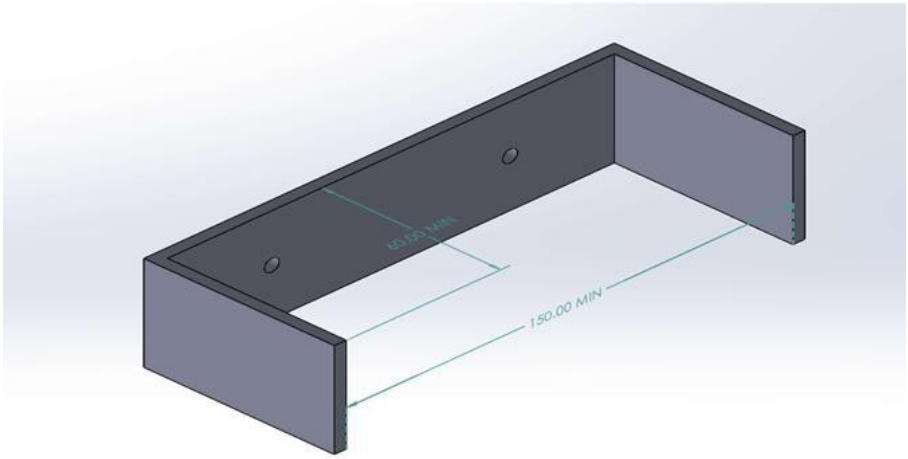
It is a competitive robotics event where manually controlled bots (wired or wireless) compete in a high-energy soccer-style match, aiming to score goals by pushing a ball into the opponent's goal while defending their own. The objective is to score more goals than the opponent before the time runs out.

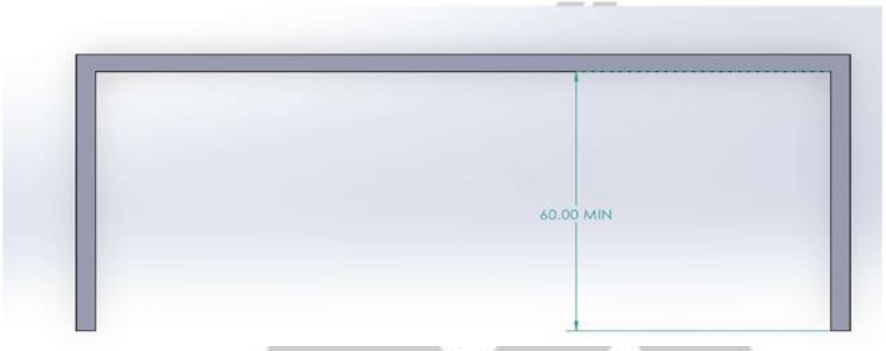
EVENT STRUCTURE:

1. Participants have to build a single robot.
2. Robots will be tested for dimensions and safety before each match.
3. Violations of rules will lead to disqualification of the respective team.
4. The winners of the qualifier matches will be declared as per the points they would have achieved in the matches.
5. **Match Duration:** Each match consists of two halves, each lasting 2 minutes and the winner will be decided based on winning criteria.
6. **Halftime:** Teams switch sides after the first half.
7. Points will be deducted in the case of multiple operators in a single match.

ROBOT SPECIFICATIONS:

1. Robot Weight:
   The maximum allowed weight of the robot is 4 kg.
2. Robot Dimensions:
   A robot should fit into a box of dimensions.
   300 mm x 300 mm x 300 mm (l x b x h) at any time during the race.
   5% tolerance is allowed in these dimensions.
3. Construction:
   No LEGO parts or ready-made assembly kits allowed.
   No kick mechanisms for ball control.
4. Clamp Specifications:
   - Minimum width: 15 cm
   - Minimum depth: 6 cm
   - Clamp must be stationary, with no moving parts.





The above images display the correct dimensions for the clamp (wedge).

5. Power Source:
   - Robots must use a fixed battery or a stationary power source connected by a cord.
   - Batteries exceeding 3S LiPo are prohibited,
   - Teams must bring spare batteries. Failure to replace a depleted battery results in disqualification.
6. Wired Robots:
   a. Wire length must be at least 2 meters and long enough to cover the arena.
   b. Wire must remain slack to prevent the movement interference.
7. Wireless robots are also allowed.

WINNING CRITERIA:

1. The team with the most goals at the end of regulation time wins.
2. The judges and event heads' decisions will be treated as final and binding to all.
3. The organizers reserve the right to change any or all of the above rules as they deem fit.

4. **Tie-breaker:** If match ends in a tie, a 1-minute tie-breaker round will be played. The first team to score wins.

ARENA SPECIFICATIONS:

Arena Size: 10ft x 6ft

Goal Posts: 500mm wide

Arena Boundaries: The field is enclosed on all sides to ensure robots stay within the playing area**.**

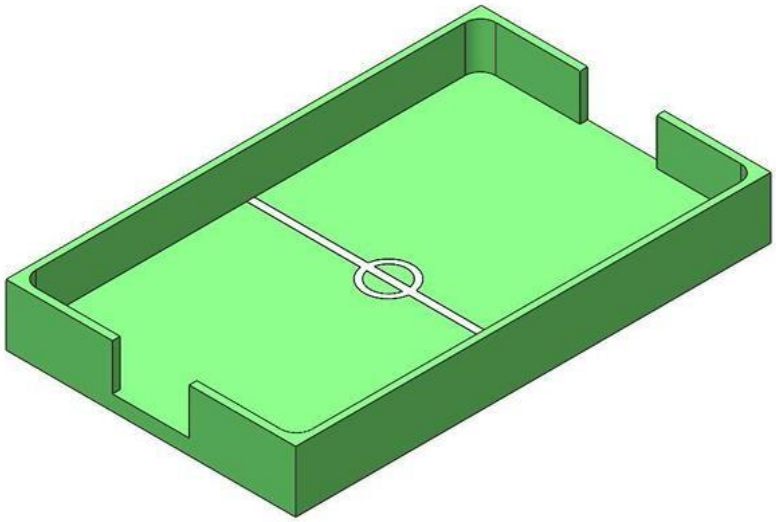
DISQUALIFICATION:

1. Violation of robot specifications or gameplay rules.
2. Deliberate interference with other robots or damage to the arena.

RULES:

1. Timely Arrival: Teams must arrive on time or face elimination.
2. Robot Handler: Only one team member can control the robot during the match. Other members must stay outside the game zone.
3. Misbehaviour: Any participant exhibiting unsportsmanlike conduct may be asked to leave and may face disqualification.
4. Decision by the judges shall be treated as final and binding to all.
5. Robot/Participant Interference:
    a. Deliberate interference with another robot results in disqualification.
    b. Any damage to the arena will lead to disqualification.
    c. Match Format: Knockout tournament format.
6. Participants are expected to maintain decorum. We reserve the due right to expel the candidate with inappropriate conduct.
7. In case of any disputes, the decision of event heads and judges will be final and binding to all.
8. Rules of the event are subject to change without prior intimation. Participants are requested to check the MindSpark'26 website (www.mindspark.org) for regular updates.
9. **Teams are advised to bring one bot per competition. If the same bot is used for multiple competitions, no rescheduling or accommodation will be provided in case of scheduling conflicts.**
10. **The same team, or the same bot, with different team name is not permitted to participate multiple times in the same event. Any violation of this rule will result in immediate disqualification.**

11. **Teams are permitted to enter the same bot in different events only if significant modifications are made to adapt the bot to the requirements of each event. For example, a bot modified by replacing its pushing edge with a C-bracket may then qualify to participate in RoboSoccer.**



PENALTIES:

1. Ramming Penalties:
   - Yellow Card: Issued after two ramming incidents.
   - 1 Point Awarded to Opponent: After three or more ramming incidents.
2. Repair Time Penalty:
   If a team fails to repair their robot within the 1- minute repair time, the opposing team receives bonus points.

TEAM AND FEE STRUCTURE:

Team Size : Maximum 4 participants per team.
Registration Fee : ₹499 per team.

EVENT HEADS:

Vedant Pawar: +91 97646 23706
Sanskar Patil: +91 75587 62021

FAQs:

Q. What is the team structure?
A. Participants can take part in a team of maximum 4.

Q. Can I participate in other events at MindSpark'26?
A. You are allowed to participate in as many events at MindSpark'26 as long as

you manage your own schedule.

NOTE:

1. No. of Prizes are subject to change depending on number of participants.
2. Registration fee is non-refundable.
3. Prize Money is subject to change.
4. Participants must report within their allotted time slot. Failure to do so will result in immediate disqualification from the event
5. The Event Head reserves the right to make the final decision in all matters concerning the event.



ROBORACES

EVENT OBJECTIVE:

This event involves the manufacturing, testing, and racing of a wireless/wired remote controlled, 2 or 4 wheeled robot.

EVENT STRUCTURE:

In this high-octane event, teams will craft and race their wireless bots through a series of thrilling rounds. Start with time trials to qualify, then challenge your rivals in a strategic elimination showdown. With a surprise track twist and fierce matchups, only the top 4 teams will make it to the finals, battling it out in a mega showdown to claim the ultimate robot-crown.

ROUNDS:-

Qualifier Round:

1. Time trials will make up this round.
2. Each team will be given a chance. The time will be recorded.
3. The first 12 teams will be chosen for the subsequent round.
4. Track for Round 1 is attached below.
5. Checkpoints will be declared during the game time.

Elimination Round:

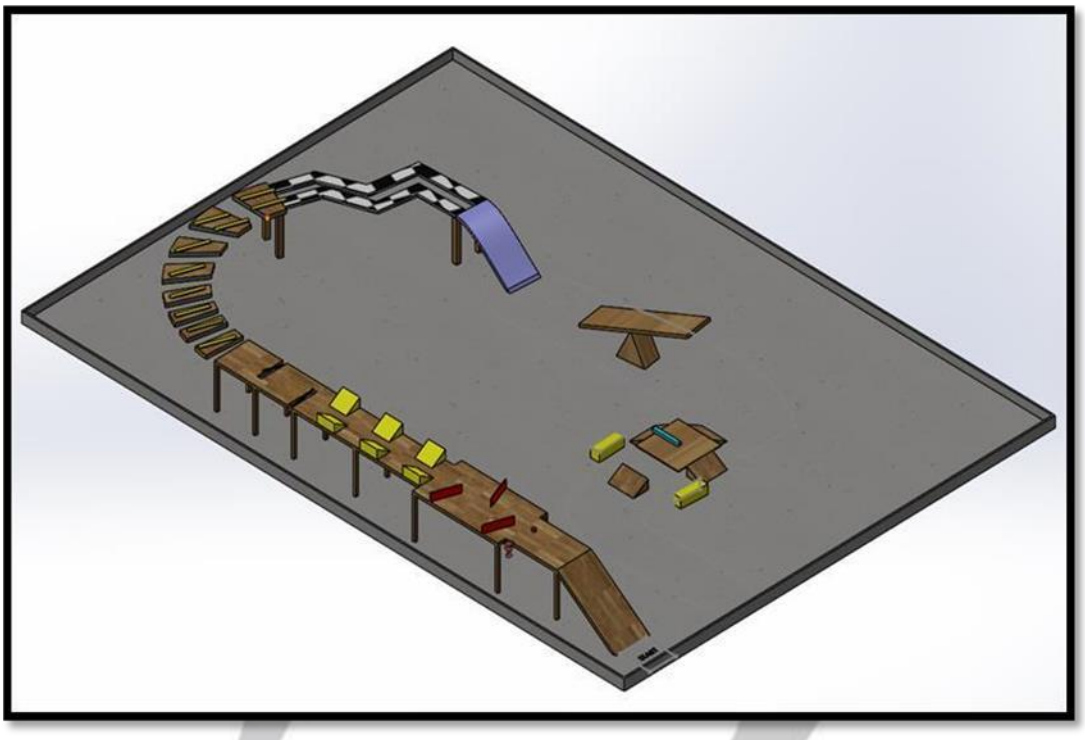
1. The top 12 teams from the first round will be selected for this round.
2. A modified track shall be disclosed at the time of the event. All teams will be given 15 minutes to analyse the track.
3. In this round, all the 12 teams race with their respective competitors, and the winning will be promoted to the next final round.
4. New Ranking will be given based on the performance in Round 2.

Finals :

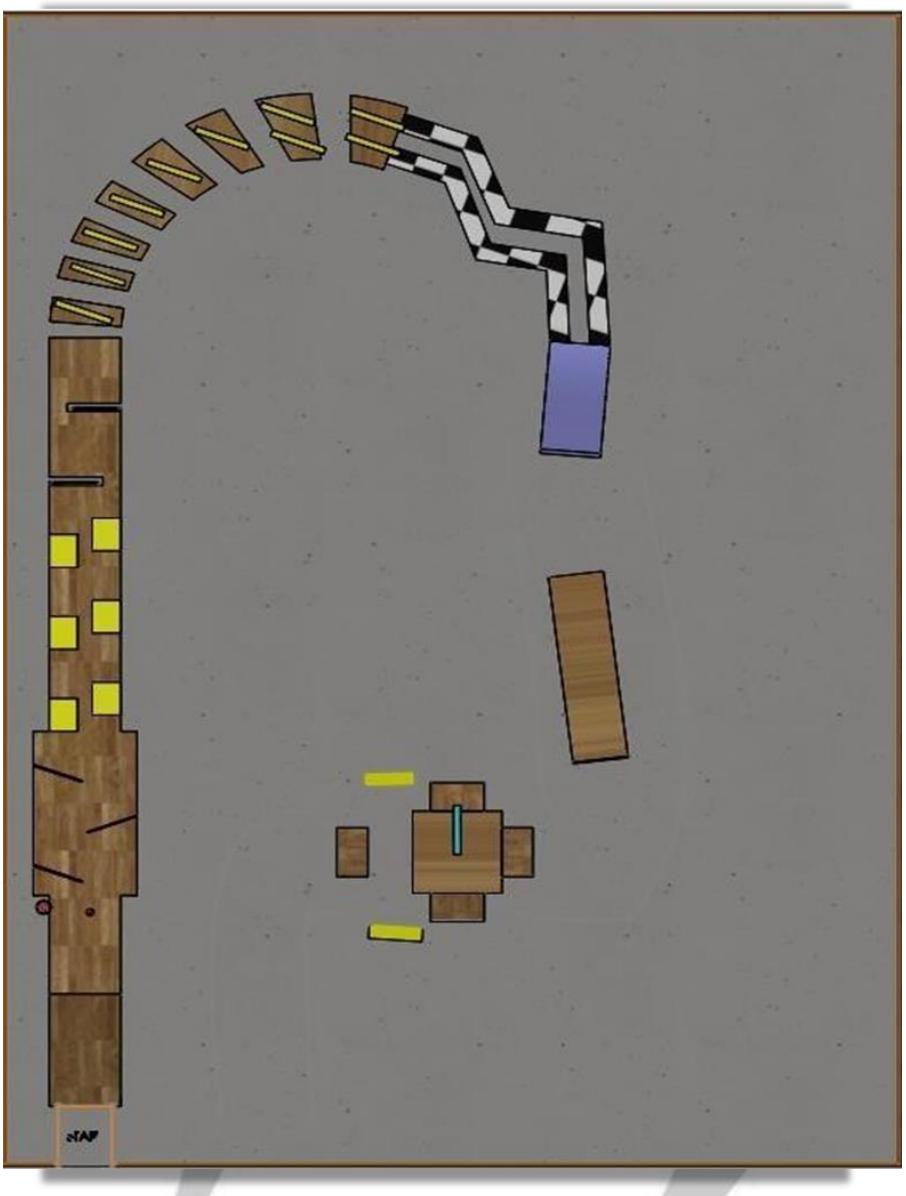
1. Final consists of individual matches. The 1v1 match format has been cancelled and will not be part of the competition.
2. Match scheduling will be decided solely by the referee. In the first-round teams will be selected as per time trails results. In later rounds matchup will be created as follows, first vs last, second vs second last, etc. Team-initiated challenges for matches are no longer applicable.
3. The four winners from above will have Individual race on the Modified track the Mega Finals. And the top three ranking will be decided.
4. The team completing the race in least time will be crowned as the winner.

ARENA:

1. The racetrack will be a combination of wedges and inclinations with many hurdles. Participants are advised to use proper motors and tyres to endure the hurdles.
2. Check points will be provided on the track where the robot can restart, in case of disturbance.



ISOMETRIC VIEW



TOP VIEW

MACHINE SPECIFICATIONS:

Battery:

Li-Po, Li-ion, or cell batteries can be used in the robot. Batteries used are to be placed in a protective shield and at the safest place on the robot so that the battery will be safe at the time of an accident.

Robot Dimensions:

A robot should fit into a box of dimensions. 300 mm x 350 mm x 350 mm (l x b x h) at any time during the race. The external device which is used to control the robot is not included in the size constraint.

Robot Weight:

The maximum allowed weight of the robot is 3 kg which includes the battery

and the remote(if wired).

Parts that can be directly procured from the market:

Motors, tires and wheels.

Power Supply:

Standard battery voltage must be 11.1V; maximum allowable voltage is 12.5V.

Radio Controllers:

Both wired and wireless connections are allowed.
Bluetooth modules, 3.0, 4.0, and 5.0 are applicable

RULES:

1. Each team can be comprised of a minimum of two members and a maximum of four members.
2. Participants must use a remote with a frequency of band spectrum 2.4 GHz only. No power supply will be given on the track. Bluetooth modules 3.0, 4.0, and 5.0 are applicable.
3. Team members are not permitted to touch either their robots or those of their opponents once the race begins (unless there is a need to lift the robot). The penalty for doing so on invalid grounds is disqualification
4. Teams are prohibited from purposefully damaging the robots of the opponent If found guilty, the concerned team will be disqualified.
5. If the robot is found unsafe to run on the track, that team will be disqualified from the race. The coordinator's decision will be final and binding to all in this No damage to the track will be tolerated, and if found guilty, the responsible team will be disqualified.
6. The robots are not allowed to leave any loose parts or any extra parts on any part of the arena.
7. Any robot that disintegrates during the race will be the vehicle must remain intact throughout the race. Once the race gets started, the participants must complete the race.
8. The track will have checkpoints at regular intervals. If the robot halts, tumbles, or goes off the arena at any point, one of the team members is allowed to lift it up and place it at the nearest completed checkpoint. However, the timer will still be running in the condition.
9. People from different institutes can be a part of the same team. However, one person cannot be a member of multiple teams at the same time.
10. Only undergraduate students may participate.

11. The judges' and organizers' decisions shall be treated as final and binding to the organizers reserve the right to change any or all of the above rules as they deem fit.
12. **Teams are advised to bring one bot per competition. If the same bot is used for multiple competitions, no rescheduling or accommodation will be provided in case of scheduling conflicts.**
13. **The same team, or the same bot, with different team name is not permitted to participate multiple times in the same event. Any violation of this rule will result in immediate disqualification.**
14. **Teams are permitted to enter the same bot in different events only if significant modifications are made to adapt the bot to the requirements of each event. For example, a bot modified by replacing its pushing edge with a C-bracket may then qualify to participate in RoboSoccer.**

TEAM AND FEE STRUCTURE:

Team size : Minimum 2 to Maximum 4 participants per team.
Registration fee : ₹ 499 per team.

EVENT HEADS:

Atharva Suryawanshi: +91 93252 30454
Gaurav Ghodam: +91 87880 66187

NOTE:

1. No. of Prizes are subject to change depending on number of participants.
2. Registration fee is non-refundable.
3. Prize Money is subject to change.
4. Participants must report within their allotted time slot. Failure to do so will result in immediate disqualification from the event.
5. The Event Head reserves the right to make the final decision in all matters concerning the event.

ROBO-ROYALE

EVENT OBJECTIVE:

A match is between red and blue teams. It last five minutes at most. Each team has one manual robot. The team which scores maximum points will win the match.

EVENT STRUCTURE:

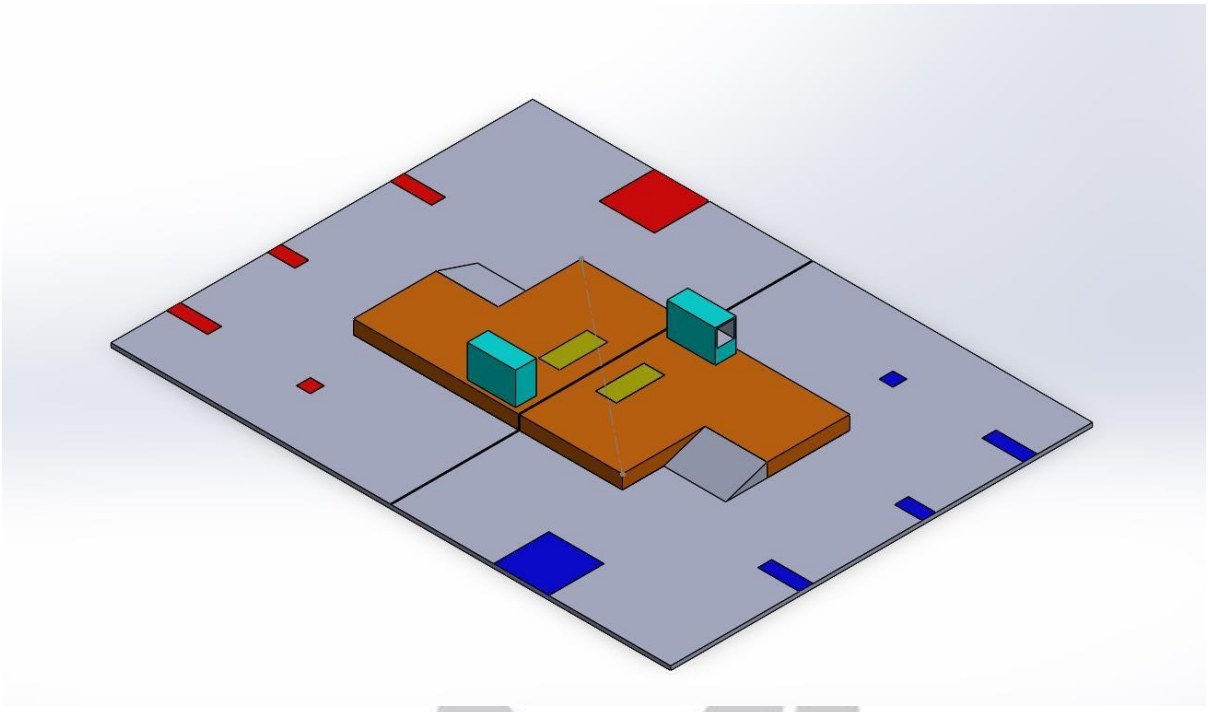
Elimination Round:

1. Each team will be given a total time of 5 minutes. They have to complete the task and score the maximum points during this round.
2. The top 8 teams will qualify for the next round.
3. The score of the team from the elimination round will not be considered in the next round.

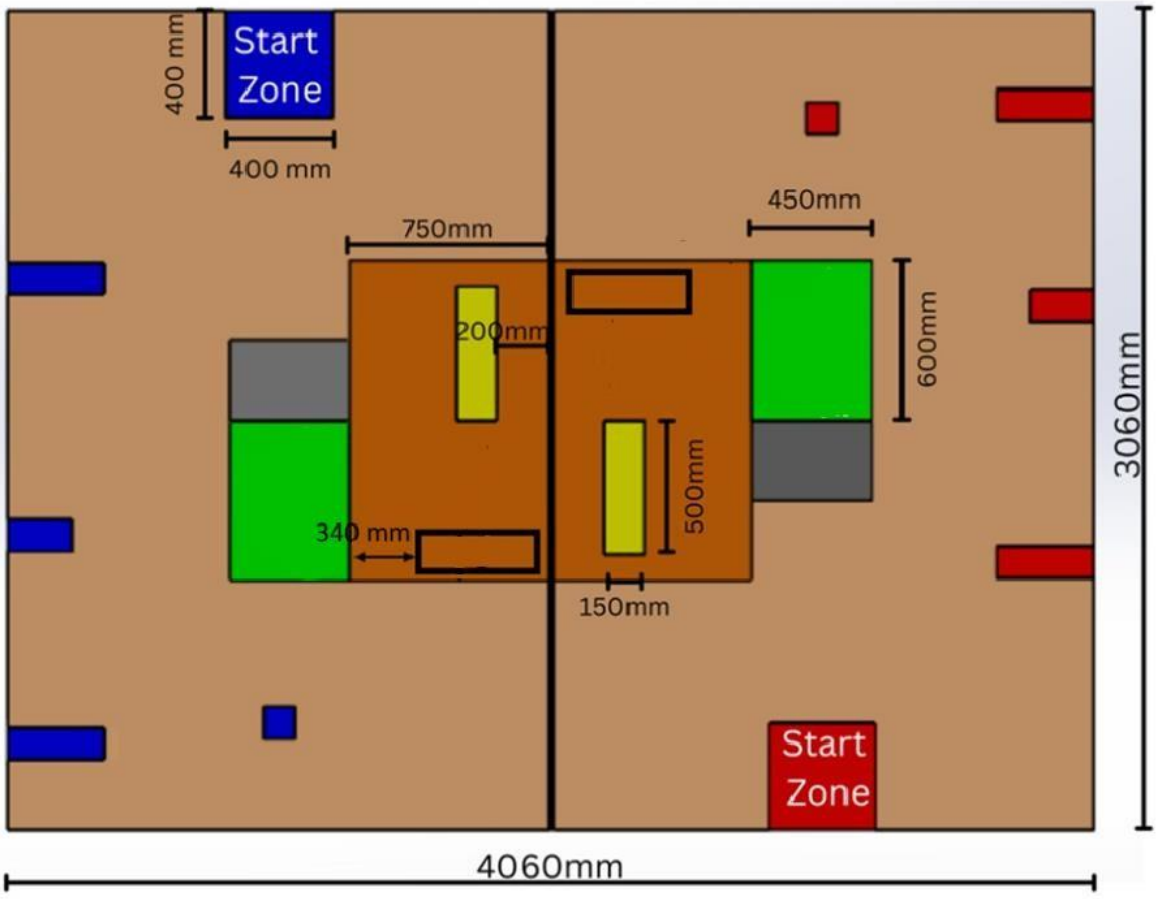
Knock-Out Round:

1. The knock-out round will start with the top 8 teams. Each match will last a maximum of 5 minutes.
2. A team, once registered, cannot register again, even with a different name.
3. A team may consist of a maximum of 5 members.
4. All students with a valid identity card from their respective educational institutes are eligible to participate.
5. Teams will be required to show the I-Cards of all of their team members during the matches.
6. A team may consist of members from different educational institutes.

ARENA DETAILS:

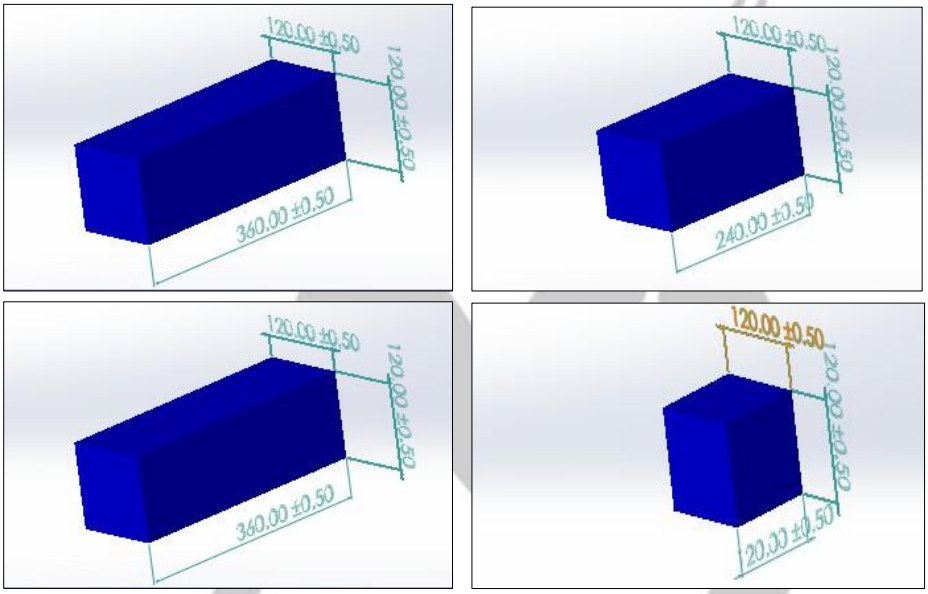


ISOMETRIC VIEW

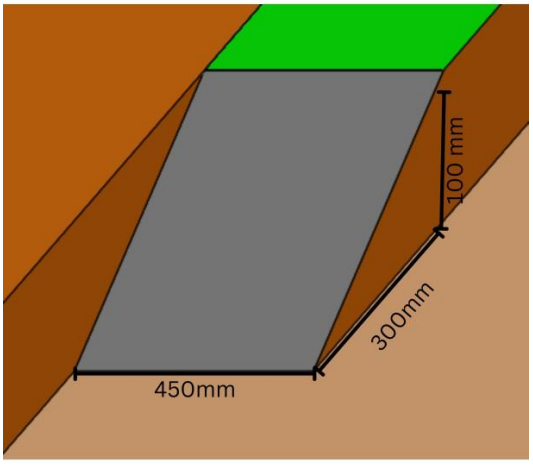


TOP VIEW

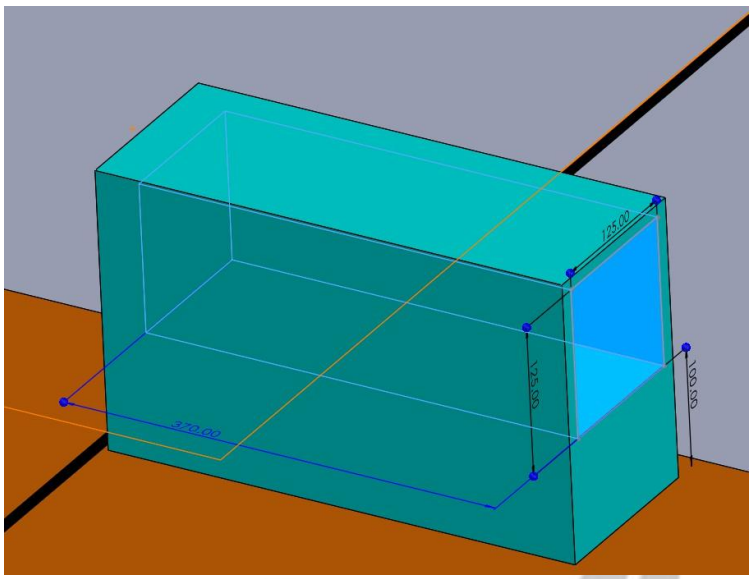
Assembly bot (AR): A Manual Robot of dimensions not exceeding 400mm x 400mm x 500mm can be controlled by a wired or wireless remote control.
Start Zone: Square of side 400mm x 400mm
Block Zone: The zone of the respective colour is where blocks are kept.
Assembly Zone: The zone where the pyramid is to be assembled is the assembly zone, and its dimensions are 150mm x 500 mm. (The yellow zone in the above images is the assembly zone).
The Blocks: The images and dimensions of the blocks used to make the pyramid are given below.



Ramp: A ramp of following dimensions is used to go towards assembly Zone.



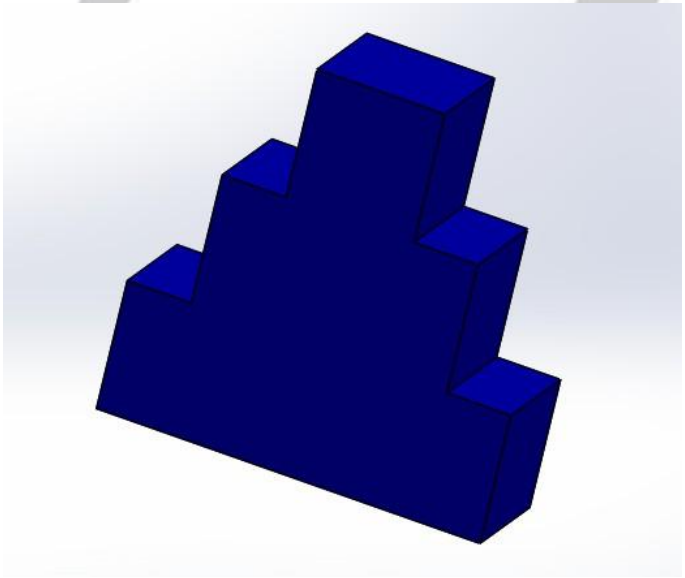
Box: Box into which the block (360±0.5) needs to be inserted.



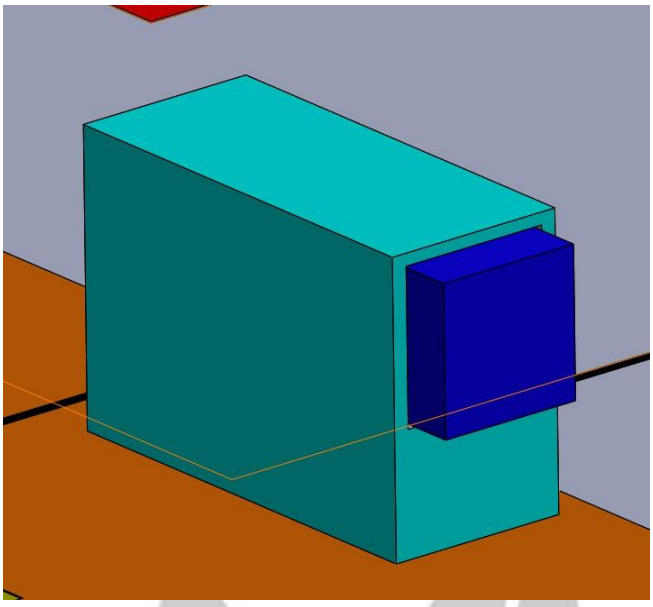
TASKS:

The assembly bot (AR) must start from Start Zone. AR bot should proceed to pick the block (one at a time).

1. **Task 1: Building a pyramid**
   The respective team bot has to pick the blocks assigned to them and assemble them in the proper order in assembly zone as shown below.



2. **Task 2: Putting block into slot**
   The team can put the 360 mm block into the slot on the box at any point of the task 1.



SCORING:

Tasks	Points
Building 1st floor	50
Building 2nd floor	70
Building 3rd floor	100
Putting Block into slot	150

MISCELLANEOUS:

1. Points will be awarded only if the block is placed correctly, i.e., it has proper orientation and is not in contact with any part of the robot.
2. If the blocks are not arranged in the proper orientation, no points will be provided for the blocks that are placed improperly.
3. Blocks piled/placed after the allotted time limit will not be given any point.
4. Points will be awarded only after a clear indication from judges.

MATCH RESULT:

1. The team with the most points will be declared the match winner.
2. In the event of a tie, the time taken to assemble the first block will be seen.

VIOLATIONS:

1. At any time, if the wire is seen to be pulling the robot, it will be considered as violation.
2. Any part of the robot extending out of the arena will lead to a violation.
3. Hand touch without prior permission by the referee will cause a violation.
4. AR running in the opponent's area will be considered a violation. Every violation will cost the team -5 points. Five points will be deducted for every violation.

MACHINE SPECIFICATIONS:

Only one AR bot is allowed. The Robot can't separate into two connected by just a wire.

1. The combined weight of the AR, including batteries and pneumatic reservoirs, should NOT exceed 10 kg.
2. The AR should fit in a box measuring 400x400x500 mm. It should fit in the AR Start zone at the beginning of the game.
3. The dimensions of the robot may change during the course of the game.
4. The AR can use an on-board power supply.
5. The cost of any damage done will have to be borne by the participants. The voltage at any point on any of the robots should not exceed 24V DC. Participants can also bring their own power supply. In this case, the maximum current and voltage limits are 5 amperes and 24 volts. The voltage and current should not exceed this limit.
6. The pressure of the pressurized non-inflammable gases in the pneumatic reservoir should not exceed 5 bars at any stage of the game.
7. The storage tank and pressure regulators used by teams need to be certified, and teams using pneumatics are required to produce the safety and security letters at the Registration Desk at the venue. Failing to do so will lead to direct disqualification.
8. Participants must be able to indicate the used pressure with an integrated or temporarily fitted pressure gauge. Also, there should be provision to check the cylinder pressure.
9. You must have a safe way of refilling the system and determining the onboard.
10. All pneumatic components on board a robot must be securely mounted. Care must be taken when mounting the pressure vessel and armour to ensure that it does not escape if ruptured.
11. The minimum length of wire extending from the robot up to the remote should be 4m. The wire should always be in slashed condition.
12. The control for the machines can be wired or wireless. In a particular match, there can be only one manual operator for the MR and one

autonomous operator for the AR. In the case of wireless controls, the participants are requested to use dual channel remotes so that there are no issues of interference during the control. The remote control and wires will not be included in the machine size.

RULES:

Retry Rules:

1. For both robots, an unlimited number of retries will be allowed.
2. The operators will first have to ask the judges for a hand-touch/retry and will take the hand-touch/retry only when allowed. The AR can restart from the start zone.
3. The orientation of the robot after the hand-touch will be decided by the judges.
4. The timer will continue to run in this period and the other team will continue the tasks.

General Rules:

1. At each hand touch, all the blocks in touch with the robot of that team (except for the ones already deposited) will be returned to their original positions.
2. Blocks will be placed in their original position after retrying if they were wrongly placed.
3. A maximum of five violations are allowed; any violations after that would lead to disqualification.
4. Any team that is not ready at the specified time will be disqualified from the competition.
5. The robot will be checked for its safety before the match and will be discarded if found unsafe for other participants, spectators, or the arena.
6. The name of your robot must be prominently displayed on the robot.
7. Operators can't step onto the arena unless permitted by the judges.
8. The AR operator has to operate from outside the arena.
9. The judge's decision shall be treated as final and binding to all.
10. The organizers reserve the right to change any or all of the above rules as they deem fit.
11. Ready-made Robots are not allowed, but ready-made sensor kits are allowed.
12. **Teams are advised to bring one bot per competition. If the same bot is used for multiple competitions, no rescheduling or accommodation**

will be provided in case of scheduling conflicts.

13. **The same team, or the same bot, with different team name is not permitted to participate multiple times in the same event. Any violation of this rule will result in immediate disqualification.**
14. **Teams are permitted to enter the same bot in different events only if significant modifications are made to adapt the bot to the requirements of each event. For example, a bot modified by replacing its pushing edge with a C-bracket may then qualify to participate in RoboSoccer.**

Safety Rules:

1. It is not permitted to use fire, water, or spray chemicals. This would lead to immediate disqualification of the team.
2. Jamming of RF control signals is NOT allowed. Teams MUST inform the judges about the RF signals involved in their robot during the robot's registration and before the start of the competition.
3. No deliberate damage must be done to the arena. The mechanism to carry objects should NOT damage them or any part of the arena. Any damage caused after prior warning has been given will lead to disqualification.
4. The decision of the Mind Spark team will be considered final in the case of any dispute.

Miscellaneous:

1. The total game field is 4060x3060 mm.
2. The game field is surrounded by walls that are 30 mm thick and 100 mm height.
3. The arena is symmetrically divided into two ideal parts for both the red and blue teams.
4. All objects and arena parts will be coated with vinyl sheets in their respective colours.
5. The thickness of all the walls is subject to change without any prior notice.
6. The colour shades of the arena and the objects are subject to variation within acceptable limits.
7. The dimensions of static objects in the arena may change subject to manufacturing difficulties.
8. 5% bidirectional tolerance is applicable to all the static objects in the arena.
9. Rules may be changed without any prior intimation. Participants are requested to check the MindSpark'26 website (www.mind-spark.org)

regularly for updates.

TEAM AND FEE STRUCTURE:

Team size : Maximum 5 participants per team.
Registration fee : ₹ 499 per team.

EVENT HEAD:

Shrisha Saundankar : +91 76664 24102
Kshitij Pawar : +91 75585 61857

NOTE:

1. Prizes are subject to change depending on number of participants.
2. Registration fee is non-refundable.
3. Prize Money is subjected to change.
4. Participants must report within their allotted time slot. Failure to do so will result in immediate disqualification from the event
5. The Event Head reserves the right to make the final decision in all matters concerning the event.

VIRTUAL ROBOTICS

EVENT OBJECTIVE:

Welcome to the Virtual Robotics Project Presentation Competition, an exciting opportunity for students to showcase their innovative projects on a virtual stage. This competition aims to celebrate technological creativity and provide a platform for students to present their ideas and designs. This is a project presentation competition that aims to provide the students with a platform to display their technical grit and receive professional feedback which shall further enhance their skills and knowledge. Virtual Robotics aims at tapping the infinite potential of the human mind to generate an idea, which, when given a proper platform, can transform into a path-breaking discovery. Robotics is often mistaken as a human replacement; nothing could be further from the truth. Robotics should be thought of more as human assistance that helps humanity unlock its full potential. In this event, you must make use of your soft skills and present your ideas to the world.

EVENT STRUCTURE:

Participants have to prepare a technical project presentation in the field of robotics, which may include concepts, designs, simulations, or research-based ideas. A working model or prototype is not compulsory but will be appreciated and add value to the presentation. Each participant or team will be allotted 10 minutes to present their idea, followed by a short Q&A session with the judges. The presentation should highlight the problem statement, methodology, innovation, applications, and future scope of the project. Evaluation will be based on presentation skills, technical depth, applicability, and understanding of the concept. The decision of the judges will be final and binding.

RULES:

1. There are no constraints on the choice of topic that is being used to present but must be technical in nature.
2. A working model of the concept is not required but appreciated.
3. The concept will be judged on various factors such as the presentation skill, applicability of the project, level of understanding of the participant, etc.
4. Participants will have to carry their own laptops compatible with VGA projector pin or necessary adapters.
5. Students will be given a time limit of ten minutes to showcase their idea. The timeline must be thoroughly adhered to.

6. A maximum of 2 participants is allowed together.
7. The decision of the judges will be final and binding. No appeals will be entertained.
8. Rules are subject to change with/without any prior notice. Participants are requested to check the official MindSpark '26 website for updates.

TEAM AND FEE STRUCTURE:

Team size : Maximum 2 participants per team.
Registration fee : ₹ 199 per team.

EVENT HEAD:

Kaveri Bramhane: +91 93097 44716
Varad Malekar: +91 86526 97100

NOTE:

1. Prizes are subject to change depending on number of participants.
2. Registration fee is non-refundable.
3. Prize Money is subject to change.
4. Participants must report within their allotted time slot. Failure to do so will result in immediate disqualification from the event
5. The Event Head reserves the right to make the final decision in all matters concerning the event.

ROBOWARS

EVENT OBJECTIVE

In this mortal combat, reach the top by being the last bot standing. Crush your opponent because only the fittest will survive.

EVENT STRUCTURE:

1. Robots will be tested for dimensions and safety before each match.
2. Violations of rules will lead to disqualification of the respective team.
3. The winners of the qualifier matches will be declared based on the points they achieve in the matches.
4. Each match will be of a duration of 5 minutes, and the winner will be decided based on the winning criteria.
5. Points will be deducted in the case of multiple operators in a single match.
6. Cluster Bots – Operator Requirement
    1. Cluster bots are permitted in the RoboWars competition.
    2. A cluster bot may consist of two or more independently mobile robots operated as a single competition entry.
    3. All robots forming a cluster bot must be controlled by a single operator using a single control system/transmitter.
    4. No additional person may directly control, operate, or provide independent commands to any individual robot within the cluster.
    5. The operator must be able to control all robots of the cluster simultaneously or through the permitted control interface.
    6. The total combined weight of all robots forming the cluster, including batteries, weapons, and all other components, must comply with the maximum weight limit specified for the respective category.
    7. All individual robots within the cluster must comply with the applicable dimensional, weapon, power, and safety regulations.
    8. The referee's decision regarding compliance with the cluster-bot rules shall be final.

RULES:

ROBOT CATEGORIES:

15 kg Category:

- Dimensions: 750mm x 750mm x 1000 mm (l x b x h) at the start of the game. Once the game starts, the robot is allowed to extend to any size.

- Weight: Must not exceed 15 kg, including the weight of the pneumatic source or tank. If the tank is external, its weight is considered 1.5 times its actual weight.

8 kg Category:

- Dimensions: 500 mm x 500 mm x 800 mm (l x b x h) at the start of the game. Once the game starts, the robot is allowed to extend to any size.
- Weight: Must not exceed 8 kg, including the weight of the pneumatic source or tank. If the tank is external, its weight is considered 1.5 times its actual weight.

Note: The weight of the remote controller and adapter will be excluded in both categories.

MOTION OF THE ROBOT:

1. Jumping and hopping are not allowed.
2. Flying (using air foils, helium balloons, or ornithopters) is not allowed.

CONTROL OF ROBOT:

1. The robot must be controlled wirelessly, with all power supplies on board.
2. Off-board power supplies are not allowed.
3. In the case of wireless control, frequency interference with other teams will not be considered for rematch or results.
4. There should be binding capability between transmitters and receivers, and they must be able to connect through polycarbonate (10 mm), metal bars, and barriers. Only remotes with such a facility will be allowed.

BATTERY AND POWER:

1. The machine can be powered electrically. The use of an IC engine in any form is not allowed. On-board batteries must be sealed, immobilized-electrolyte types (such as gel cells, lithium, NiCad, NiMH, or dry cells).
2. The electric voltage between 2 points anywhere in the machine should not exceed 36 V DC at any point. If a team uses AC voltage in any part of the machine, it should not exceed 36 V AC.
3. All efforts must be made to protect battery terminals from a direct short and prevent a battery fire. Failure to do so will cause direct disqualification.
4. Use of damaged, non-leak-proof batteries may lead to disqualification.
5. Special care should be taken to protect the on-board batteries.

HYDRAULICS:

1. A robot can use non-inflammable liquid to actuate hydraulic devices
2. All hydraulic components on board must be securely mounted.
3. Special care must be taken while mounting the pump, accumulator, and

armour to ensure that if ruptured, direct fluid streams will not escape the robot.

4. The whole hydraulic system should be leak-proof. The maximum allowed pressure is 8 bars.
5. Participants must be able to indicate the used pressure with an integrated or temporarily fitted pressure gauge.

PNEUMATICS:

1. A robot can use pressurized non-inflammable gases to actuate pneumatic devices. The maximum allowed outlet pressure is 8 bars. The storage tank and pressure regulators used by the teams need to be certified. Teams using pneumatics are required to produce the safety and security letters at the registration desk at the venue. Failing to do so will lead to direct disqualification.
2. Participants must be able to indicate the used pressure with an integrated or temporarily fitted pressure gauge. There should be a provision to check the cylinder pressure on the bot.
3. A safe way of refilling the system and determining the on-board pressure must be provided. All pneumatic components on board must be securely mounted. Care must be taken while mounting the pressure vessel and armour to ensure that if ruptured, it will not escape the robot. Failing to do so will lead to direct disqualification.

WEAPONS:

Robots can have any kind of cutters, flippers, saws, lifting devices, spinning hammers, etc., as weapons. Exceptions and limitations for weapons include:

1. Liquid projectiles.
2. Any kind of inflammable liquid.
3. Flame-based weapons.
4. Any kind of explosive or intentionally ignitable solid.
5. Nets, tape, glue, chains, or any other method of entanglement.
6. High-power magnets or electromagnets.
7. Radio jamming, lasers, tesla coils, or any other high-voltage devices.
8. Tethered or untethered projectiles.
9. Spinning weapons that do not come into contact with the arena at any point are allowed. However, the arena should not be damaged by any bot.

SAFETY RULES:

1. Special care should be taken to protect the on-board batteries and pneumatics. Robots without proper protection will not be allowed to compete.

2. Robots must be activated in the arena or with the expressed consent of the event coordinators.
3. All weapons must have a safety cover on any sharp edges.
4. All participants build and operate robots at their own risk. Please take care to avoid injuring yourself or others when building, testing, and competing. Organizers will not be responsible for any injuries to operators or team members.
5. In the event of any disputes, the decision of the organizers and judges will be final and binding on all.
6. Rules are subject to change with/without any prior notice. Participants are requested to check the official MindSpark '26 website for updates.

WINNING CRITERIA:

1. A robot is declared victorious if its opponent is immobilized.
2. A robot will be declared immobile if it cannot show linear or rotational motion of at least one foot within 30 seconds. A robot with one side of its drive train disabled will not be counted as immobilized if it can demonstrate some degree of controlled movement. If both robots remain mobile after the end of the round, the winner will be decided subjectively. A robot deemed unsafe by the judges after the match has begun will be disqualified and declared the loser. The match will be immediately halted, and the opponent will be awarded a win.
3. The judges' and event heads' decisions will be treated as final and binding to all.
4. The organizers reserve the right to change any or all of the above rules as they deem fit.
5. In the event of a match resulting in a tie, a rematch of 2 minutes will be played. If a further tie occurs, the team scoring the highest points in the first 3 minutes of the main match will be declared the winner.
6. Points will be given based on aggression, damage, control, and strategy.
    a) **Aggression (3 points):** Judged by the frequency, severity, boldness, and effectiveness of attacks deliberately initiated by the robot against its opponent. Accidental attacks will not be considered Aggression.
    b) **Control (3 points):** A robot is considered controlled if it can attack an opponent at its weakest point, use its weapons effectively, avoid Arena Hazards, and minimize damage from the opponent.
    c) **Damage (5 points):** Damage is scored when a robot, either directly or indirectly using Arena Hazards, reduces the functionality, effectiveness, or defensibility of an opponent. Self-inflicted damage is not counted.

ARENA SPECIFICATIONS:

1. **Arena Size:** 17 x 17 x 8 ft
2. **Operating Area:** 16 x 16 ft
3. **Polycarbonate Covering:** 10 mm Lexan on all sides
4. **Floor Thickness:** MS 4 mm

TEAM AND FEE STRUCTURE:

Team Size : Maximum 5 participants per team.
Registration Fee : ₹599 per team

EVENT HEADS:

Shreyas Narade: +91 96994 43159
Suparshv Patil: +91 96372 51008

FAQs:

Q. What is the team structure?
A. Participants can take part in a team of maximum 5.

Q. Can I participate in other events at MindSpark'26?
A. You are allowed to participate in as many events at MindSpark'26 as long as you manage your own schedule.

NOTE:

1. No. of Prizes are subject to change depending on number of participants.
2. Registration fee is non-refundable.
3. Prize Money is subject to change.
4. Participants must report within their allotted time slot. Failure to do so will result in immediate disqualification from the event.
5. The Event Head reserves the right to make the final decision in all matters concerning the event.

SEARCH N' DESTROY

EVENT OBJECTIVE.

Search N' Destroy is basically a line-following and maze competition. A track of white lines on black backgrounds and black lines on white backgrounds will be laid out and an autonomous robot must follow that line. The robot must complete a task. The path will have checkpoints. Each checkpoint will earn some points based on the number of checkpoints covered; the time taken to cover these checkpoints as well as the task completed will decide the points earned by the robot. The team which follows the correct path the fastest will be the winner.

EVENT STRUCTURE:

Arena:

1. All dimensions are in mm.
2. The maximum size of the arena will be 5200mm x 4500mm
3. The width of the track will be 30mm.
4. The dimensions of the start zone are 200mm x 200mm.
5. The floor arena shall be made of wood (plywood) and finished by vinyl. Do not assume that the floor provides the necessary amount of friction.
6. Participants should not assume that the track is always black, as fading may occur.
7. The track will have discontinuities at some.
8. All the possible checkpoints are marked as black stripes perpendicular to the track having the dimensions of 90mm x 30 mm. The checkpoints are CP1, CP2 & CP3.
9. The dimensions of the arena will be accurate to within 5%, or 20mm.
10. Do not make any assumptions about the amount of sunlight, incandescent light, or fluorescent light that may be present at the contest.
11. A cube of dimension 100mm x 100mm x 100 mm is kept in the yellow zone.
12. The arena is divided into two sections for each team by a central wall, which will be constructed using coloured plywood (red/blue).

MACHINE SPECIFICATIONS:

Dimensions and Fabrications:

1. The machine should fit within a red-colored square measuring 200mm x 200 mm in the figure. There are no height restrictions.
2. The autonomous robot must be stable and capable of standing on its own at the beginning of the run when placed in the starting zone. Robots that do not meet this criterion will be disqualified.

3. During the run, the autonomous robot can expand itself if it does not damage the arena in any way. However, the machine should not leave anything behind while tracing the path. The machine should not scratch, damage, or destroy the track or any accompanying parts of the arena.
4. If any robot is found damaging the arena, the final decision rests at the discretion of the organizers.
5. Teams are permitted to use pre-made sensor kits. However, they are not allowed to use Lego kits or any other similar assemblies.

Power Supply and Propulsion:

1. The machine must be entirely self-contained and should not receive any external assistance. It must not utilize an energy source that involves a combustion process.
2. The machine must have its own on-board power source. The use of external power supplies is not permitted.
3. The potential difference between any two points on the machine must not surpass 24V.

Tasks:

1. Teams are required to construct an autonomous robot capable of line following.
2. All minimum possible paths will comprise of 3 checkpoints and a finishing zone.
3. The robot has the option to either follow the provided path or select its own route, with a preference for adhering to the given path.
4. The task will conclude when the cube (initially positioned in the yellow zone) is relocated to the green zone located behind the opposing team's red zone.

Note:

In the subsequent rules, the action of the robot pushing the box and causing it to come to rest in the green zone is denoted as "Destroying the Opponents' Territory." The robot must refrain from causing any physical harm to the walls or the cube.

RULES:

Competition:

1. Each participating team will be granted a 2-minute dry run session on the arena.
2. During the dry run, teams can calibrate sensors and explore the entire arena to identify the desired path.
3. Subsequent to the dry run, robot programming is prohibited.

4. The bot's starting procedure must be straightforward and devoid of any manual force or directional impulse.
5. The game's duration is 5 minutes. The runtime clock initiates after 2 minutes of the dry run and concludes after 3 minutes of gameplay, even if sensor adjustments or checkpoint searches are ongoing.
6. The match initiates from the start zone, and the operator retains the authority to abort the run at any juncture.
7. Only a single team member, namely the operator, can enter the arena after securing approval from the organizers.
8. Operator area will be allotted and operator will not be allowed on the arena till the referee allows.
9. The robot is permitted to push the cube only after successfully passing all checkpoints.
10. The robot must avoid causing any physical damage to the walls or the cube.

Checkpoints:

As depicted, the arena will feature multiple black-colored checkpoints. To meet the requirement, the robot must pass through a minimum of 4 checkpoints, which includes crossing the finish line. A checkpoint will be deemed cleared only when the entire robot has traversed through it, signifying completion of the entire arena.

Hand Touch:

1. If an operator touches the robot during a run, it will be deemed a hand touch, and the robot must be placed back at the last restart or checkpoint it had crossed. (Robots can be removed after the completion of a match, such as after placing the cube.)
2. The operator can request prior permission from the organizers to perform any number of hand touches. There is no penalty for a hand touch with prior permission; however, the robot must be returned to the start or the last checkpoint crossed.
3. Even if a participant executes a hand touch, the clock will continue to run and will not be reset, as the robot resumes from the previous restart point.
4. In case a robot encounters difficulties, the operator can seek permission from the judge to perform a hand touch and restart the robot from the start or the last checkpoint. If the operator touches the robot without the prior permission of the judge during run time, a deduction of 10 points will be imposed.

General Rules:

1. Teams not ready at the specified time will be automatically disqualified from the competition.
2. Prior to the match, the machine will undergo a safety check, and if found

unsafe for other participants and spectators, it will be discarded.

3. The machine must not leave any marks on the floor; any robot causing damage to the arena will be immediately disqualified.
4. Only one team member is permitted to handle the robot.
5. Participants are restricted from placing anything inside the arena except for their autonomous robots.
6. We will not provide any type of computer, software, laptop, programmer, or IC under any circumstances.
7. Laptops or personal computers are prohibited near the arena. Other devices like Wi-Fi, Bluetooth, etc., must be turned off. Organizers have the authority to inspect these devices and their usage.
8. No robot is allowed to employ flammable, combustible, explosive, or potentially hazardous processes.
9. The time recorded by the organizers will be deemed final and used for scoring the teams. Any time measured by a contestant using other means is not acceptable for scoring.
10. A maximum score of 4 checkpoints will be awarded.
11. The operator must wear socks.
12. Rules are subject to change with/without any prior notice. Participants are requested to check the official MindSpark '26 website for updates.

JUDGING CRITERIA:

Scoring System:

1. Each checkpoint carries 20 points.
2. Placing the box in the green zone carries 40 points.
   Maximum points = 100.

In the event of a tie in scores between two teams, preference will be determined based on the following criteria in the specified sequence:

1. The shortest time taken by robots to cross the same number of checkpoints.
2. The fewest number of hand touches.
3. To be decided by the organizers

TEAM SPECIFICATIONS:

1. Once registered, a team cannot register again, even under a different name.
2. A team may comprise a maximum of 5 members.

TEAM FEE AND STRUCTURE:

Team size : Maximum 5 participants per team
Registration fee : ₹ 499 per team.

EVENT HEADS:

Pratik Suryawanshi: +91 70352 10033

Sumit Sonawane: +91 74990 51061

NOTE:

1. Power supply will not be available during the match, but it can be provided.
2. Prizes are subject to change based on the number of participants.
3. The registration fee is non-refundable.
4. Participants must report within their allotted time slot. Failure to do so will result in immediate disqualification from the event.
5. The Event Head reserves the right to make the final decision in all matters concerning the event.